A Tutorial on Stance Detection

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ABSTRACT

Stance detection (also known as stance classification, stance prediction, and stance analysis) is a problem related to social media analysis, natural language processing, and information retrieval, which aims to determine the position of a person from a piece of text they produce, towards a target (a concept, idea, event, etc.) either explicitly specified in the text, or implied only. Common stance classes include *Favor*, *Against*, and *None*. In this tutorial, we will define the core concepts and other related research problems, present historical and contemporary approaches to stance detection (including shared tasks and tools employed), provide pointers to related datasets, and cover open research directions and application areas of stance detection. As solutions to stance detection can contribute to diverse applications including trend analysis, opinion surveys, user reviews, personalization, and predictions for referendums and elections, it will continue to stand as an important research problem, mostly on textual content currently, and particularly on Web content including social media.

CCS CONCEPTS

• **Computing methodologies** → **Natural language processing**; **Machine learning**; **Language resources**; • **Information systems** → **Information retrieval**; **Web and social media search**; **Sentiment analysis**.

KEYWORDS

Stance detection, Social media analysis, Twitter, Data streams

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1 INTRODUCTION

Stance detection has recently emerged as an important research problem in natural language processing (NLP), social media analysis, and information retrieval (IR), as revealed in the number of related papers published particularly since 2015 [15]. It is a research topic considered closely related to sentiment analysis and is commonly defined as the detection of the stance (as *Favor*, *Against*, or *None*) of the text producer towards a target [22, 23]. Apart from sentiment analysis, there are other research problems closely related to stance detection, including emotion detection, sarcasm detection, irony detection, controversy detection, and argument mining, among others.

Stance detection is known to have several application areas such as prediction of election/referendum results, information retrieval, rumour classification, and fake news detection. Particularly based on the latter two application areas, two subproblems of stance detection have emerged, namely, *rumour stance detection* and *fake news stance detection* [15]. Definitions of these subproblems along others (*multi-target stance detection* [27] and *cross-target stance detection* [39]) will be provided within the course of this tutorial.

An important contribution to the stance detection research is a series of shared tasks on stance detection between 2016 and 2017: an initial competition (within the course of SemEval-2016) on English tweets [22], another on Chinese microblog posts [40], and another related competition on Spanish and Catalan tweets [31]. More recently, a stance detection shared task has also been performed for tweets in Italian [6]. Another related shared task on Spanish and Basque tweets is again conducted recently [18]. In addition to the evaluation of different approaches for stance detection on microblogs in different languages, these competitions have facilitated the compilation significant annotated datasets [6, 18, 21, 22, 40] for stance detection.

Earlier work on stance detection utilized different rule-based and machine learning based algorithms on a variety of text genres including congressional-floor debates, online debate forums, student essays, and tweets [8, 9, 25, 32]. Related studies seem to have increased noticeably after the aforementioned initial shared task on stance detection in English tweets [22]. Hence, particularly considering those studies after (and including) this shared task; traditional machine learning approaches like SVM [1, 11, 12, 22, 23, 26, 38] and logistic regression [6, 24, 41, 42], deep learning approaches like LSTM [2, 7, 28, 29, 36] and CNN [35, 37, 43], and ensemble methods [20, 30, 33] have all been employed for the task of stance detection. Though machine learning approaches like SVM are the most commonly utilized ones up until 2019, more recent studies tend to apply deep learning algorithms, similar to the case of many tasks in NLP and IR [15]. There also exist significant research efforts to produce annotated datasets for stance detection [14, 17, 27, 44]. Currently, stance-annotated datasets exist for a number of languages including English [21, 27], Catalan [31], Chinese [40], Czech [10], Italian [19], Spanish [31], Turkish [13, 14], and Basque [18]. Yet, in addition to these language-specific datasets, recent work on the compilation of multilingual stance-annotated datasets [17, 34, 44] (where these datasets include annotated samples in other languages such as German and French) will facilitate related research and thereby will hopefully lead to important findings. There are several lines

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of future research after having a firm base knowledge about the stance detection problem. Examples include cross-lingual and multilingual stance detection [15], stance detection in data streams [3], stance detection on non-textual content, context-sensitive stance detection [15], and exploring the application opportunities of new methods [4, 5].

This tutorial aims to cover the core concepts in stance detection together with related research topics, significant approaches to this problem published so far, relevant datasets and other practical resources, application areas, and finally, outstanding issues not sufficiently addressed in the related literature so far. We believe that this tutorial will be beneficial to interested researchers of Web search, NLP and IR alike, as well as related practitioners.

2 INTENDED AUDIENCE

Stance detection is an important research problem within the domains of social media analysis, Web search, natural language processing, and information retrieval. Therefore, we believe that researchers and practitioners of these research topics can readily benefit from this tutorial. There are not any particular prerequisites for the attendees.

3 OBJECTIVES OF THE TUTORIAL

The objective of this tutorial is to provide an overall understanding of the stance detection problem, which has significant and widespread application opportunities. The tutorial attendees will:

- grasp the main stance detection problem, its subproblems, and its relationship with other problems closely related to stance detection,
- learn about the evolution of the stance detection literature, and about common algorithms and approaches to this problem, in addition to the tools employed,
- be provided with pointers to related common stance detection datasets so that they can readily begin conducting stance detection experiments themselves on the available datasets, and
- learn about the related application areas as well as open research topics regarding stance detection, and learn about some common and favorable practices employed by stance detection researchers and practitioners.

4 PREVIOUS STANCE DETECTION TUTORIALS

To the best of our knowledge, there were two previous tutorials on stance detection. The first one was titled "*Detection and Characterization of Stance on Social Media*" which was carried out within the course of *14th International Conference on Web and Social Media (ICWSM-2020)*. There are several differences between our tutorial on stance detection and this previous tutorial: (1) Previous tutorial focuses on social media only while our tutorial will have a broader coverage including other input genres as in fake news stance detection where news articles are under consideration. (2) Being a more recent tutorial on the topic, our tutorial will cover more recent work which is a significant advantage as considerable body of work is being published on stance detection each year. (3) In our tutorial, we will pay particular attention to the subproblems of stance detection and other closely-related problems, in addition to significant application areas and outstanding issues.

The second tutorial on stance detection was our tutorial titled "*Stance Detection: Concepts, Approaches, Resources and Outstanding Issues*" which was carried out within the course of *44th International ACM SIGIR Conference on Research and Development in Information Retrieval (SIGIR-2021)* [16]. Our current tutorial proposal will cover more recent work on stance detection, compared to the content of our previous tutorial. For instance, a new shared task named VaxxStance-2021 [18] has been performed in 2021 and work related to this recent shared task will be included in our tutorial. Similarly, tweet datasets on stance detection for new languages such as Basque [18] are being published within the course of more recent work. In this tutorial at SIGIR-2021, there were questions from the audience regarding the differences and interrelationships between stance detection and related problems such as sentiment analysis and controversy detection. We addressed these questions according to the findings of the related work [15, 23]. There were also questions regarding the annotation procedures of stance detection datasets and these questions were also answered based on the related previous work on stance detection and related problems. The audience was provided with pointers to relevant literature on stance detection.

5 TUTORIAL OUTLINE

The outline of our tutorial on stance detection is presented below.

(1) Introduction
(2) Core concepts and related problems
    - Definitions of stance detection and subproblems of stance detection.
    - Interrelationships with related problems including sentiment analysis, emotion recognition, sarcasm/irony detection, controversy detection, and argument mining
(3) Stance detection competitions (shared tasks)
    - SemEval-2016 Task 6: Detecting Stance in Tweets
    - Shared Task of Stance Detection in Chinese Microblogs at NLPCC-ICCPOL-2016
    - Shared Task of Stance Detection in Spanish and Catalan Tweets at IberEval-2017
    - SardiStance: Stance Detection Task in Italian Tweets at EVALITA-2020
    - VaxxStance: Going Beyond Text in Cross-lingual Stance Detection at IberLeF-2021
(4) Historical and contemporary approaches
    - Earlier studies on stance detection
    - Machine learning based approaches
    - Deep learning based approaches
    - Ensemble learning approaches
    - Related learning systems and tools
(5) Common stance detection datasets
(6) Application areas
    - Opinion surveys/polling
    - Public health surveillance
    - Information retrieval
    - Stance summarization
    - Rumour classification

- Fake news detection

(7) Outstanding issues
- Stance detection in data streams
- Cross-lingual and multilingual stance detection
- Context-sensitive stance detection

(8) Conclusion

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